

in specialised labs to catch EMI/EMC and safety issues. Regulatory submissions demand thorough documentation and test evidence. Each technical requirement should be linked to a corresponding test case, ensuring traceable and verifiable compliance throughout the product lifecycle.

### Embedded system and IoT gateway design: A practical example

Designing a reliable embedded system and IoT gateway begins with a clear definition of the system scope and mission—what the device is intended to do and the environment it will operate in. Choosing the right microcontroller and hardware components is critical, with emphasis on reliability, power efficiency, and built-in security features. The firmware architecture should be modular and maintainable, with support for over-the-air (OTA) updates to enable future enhancements and bug fixes. Hardware-software co-design plays a vital role, requiring close collaboration between hardware and firmware teams to achieve seamless integration, optimal performance, and efficient development cycles.

The IoT gateway serves as the bridge between embedded devices and cloud or enterprise systems. On the hardware side, it must have sufficient processing power and memory to handle data aggregation, edge analytics, and protocol translation in real time. Multi-protocol connectivity is a must—supporting Ethernet, Wi-Fi, LTE, RS485, CAN, BLE, Modbus, OPC UA, and more—to ensure compatibility with diverse devices and networks. Industrial use cases demand ruggedisation against environmental factors like temperature extremes, humidity, vibration, and dust ingress, with protection ratings such as IP54, IP65, or IP68. Robust power management options, including battery, mains, or solar sources with backup systems and low-power modes, further enhance reliability in the field.

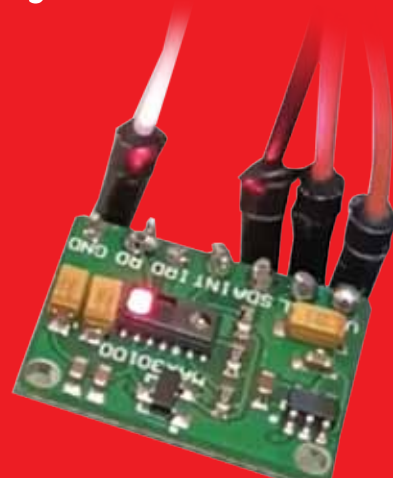
From the software and security perspective, the gateway may run a real-time operating system (RTOS) or embedded Linux, depending on the complexity and latency requirements of the application. Edge computing capabilities allow for local data processing, filtering, and analytics, reducing dependency on cloud bandwidth. Security is foundational—requiring encrypted communication, secure boot, hardware-based root of trust, and role-based access control to protect against cyber threats.

For integration and interoperability, the gateway must support protocol translation, enabling seamless communication between legacy systems and modern IoT devices. A modular architecture based on APIs and microservices ensures scalability and simplifies integration with other systems. Remote management

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